

Inflammatory Chemokine Activation and Cell State Continua in Human Intervertebral Disc Degeneration: A 12-Dataset Single-Cell Transcriptomic Meta-Analysis

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Companion Analysis: An independent analysis of 7 of these 12 datasets (173,628 cells, 29 donors) was performed using a separate pipeline (Harmony integration, R-based DESeq2, LIANA CCC). That analysis and its full report are available at [phylo_analysis/report_IVD_scRNAseq_analysis.md](#), with a corresponding draft manuscript at [phylo_analysis/draft_manuscript.md](#). The phylo analysis is referenced throughout this document where its findings converge with or diverge from the present 12-dataset analysis.

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1. Abstract

Intervertebral disc (IVD) degeneration is the primary structural cause of chronic low back pain, affecting over 600 million people worldwide (GBD 2021 Low Back Pain Collaborators, 2023). To comprehensively map its cellular and molecular landscape, we integrated 12 publicly available human scRNA-seq datasets comprising 436,239 cells from 78 samples (57 donors) across nucleus pulposus (NP), annulus fibrosus (AF), and cartilage endplate (CEP) compartments. Using a tiered integration strategy (scANVI for resident cells, scVI for

non-resident cells), we identified cell types existing on a continuum from notochordal to mature chondrocyte to stressed/degenerative states. Pseudobulk differential expression with pyDESeq2 identified 5,328 significant genes across 17 powered comparisons, revealing a classical inflammatory/catabolic signature dominated by CXC chemokines (CXCL1 $\log_2FC=+3.75$, CXCL3 $+3.72$, CXCL2 $+3.13$) and TNF ($+2.45$) in severe NP degeneration. Pathway enrichment confirmed chemokine-mediated signaling, neutrophil chemotaxis, and inflammatory response as the dominant upregulated programs in NP cells, alongside heat shock protein activation and mitochondrial dysfunction in AF cells. Transcription factor analysis identified RELA/NFKB1, HSF1/HSF2, and E2F4 as key regulators. PAGA/diffusion pseudotime trajectory analysis demonstrated that pseudotime correlates with disease condition (NP $\rho=-0.207$, AF $\rho=-0.177$, both $p<10^{-100}$), with ~55% overlap between trajectory-associated and DE genes. Cell-cell communication analysis (LIANA) revealed increased signaling complexity in degeneration (53,036 vs 44,079 interactions). Pain gene analysis showed that disc cells produce inflammatory mediators (TNF, CXCL1-3, PTGS2) but not nociceptors, consistent with a model of inflammation-driven nerve sensitization rather than direct pain signaling. These findings define an inflammatory chemokine-driven mechanism of IVD degeneration and identify TNF/NF- κ B inhibition, CXC chemokine blockade, and heat shock protein modulation as candidate therapeutic strategies.

2. Introduction

2.1 The Clinical Problem

Low back pain (LBP) is the leading cause of years lived with disability worldwide, affecting approximately 619 million people and imposing annual costs exceeding \$100 billion in the United States alone (GBD 2021 Low Back Pain Collaborators, 2023; Dieleman et al., 2020). Approximately 40% of symptomatic LBP is attributable to IVD degeneration (Wang et al., 2023a). Despite decades of research, no disease-modifying therapy exists; current treatments are limited to symptomatic relief (analgesics, physical therapy) or surgical intervention (discectomy, fusion) for refractory cases.

2.2 IVD Structure and Biology

The IVD is a fibrocartilaginous structure comprising three compartments: the nucleus pulposus (NP), a highly hydrated gel-like core rich in aggrecan and type II collagen that absorbs compressive loads; the annulus fibrosus (AF), a tough outer ring of concentric collagen I-rich lamellae providing tensile strength; and the cartilage endplate (CEP), thin hyaline cartilage layers that serve as the primary route for nutrient diffusion into the avascular NP (Oichi et al., 2020). The NP is the largest avascular structure in the human body, forcing its resident cells to operate under near-anoxic conditions via anaerobic glycolysis (Oichi et al., 2020).

2.3 Pathomechanisms of Degeneration

IVD degeneration is characterized by progressive loss of NP hydration through aggrecan degradation by ADAMTS4/5 and MMPs (Liang et al., 2022), inflammatory activation driven by TNF- α , IL-1 β , and NF- κ B signaling (Risbud and Shapiro, 2014; Xia et al., 2024), cellular senescence and apoptosis (Song et al., 2023a), oxidative stress from mitochondrial dysfunction (Song et al., 2023b; Wang et al., 2023a), and fibrocartilaginous replacement of the NP by type I collagen-producing cells (Antoniou et al., 1996). In advanced degeneration, nerve fibers and blood vessels invade the normally avascular NP through AF fissures, contributing to discogenic pain (Freemont et al., 2002).

2.4 Rationale for Single-Cell Meta-Analysis

Prior single-cell studies of the IVD have been limited by small sample sizes (2-7 donors), single datasets, or focus on a single compartment (Gan et al., 2021; Fernandes et al., 2020; Li et al., 2022a). By integrating 12 datasets spanning 57 donors, we aimed to create the most comprehensive single-cell atlas of human IVD degeneration and achieve sufficient statistical power for pseudobulk differential expression analysis, the gold standard for scRNA-seq DE that avoids the inflated false positive rates of naive single-cell approaches (Squair et al., 2021; Zimmerman et al., 2021).

A critical methodological consideration is the distinction between resident disc cells (NP, AF) and non-resident cells (immune, endothelial). IVD resident cells exist on a phenotypic continuum — from notochordal to mature chondrocyte to stressed/degenerative states — that can be erased by aggressive batch correction (Gan et al., 2021). Our tiered integration strategy addresses this by applying conservative, semi-supervised integration (scANVI) to resident cells while using standard approaches for non-resident populations.

3. Study Design and Datasets

3.1 Dataset Selection

Twelve publicly available scRNA-seq datasets of human IVD tissue were identified from GEO and CNGB. Selection criteria included: (1) human IVD tissue, (2) single-cell resolution (not single-nucleus), (3) raw count matrices available.

Table 1. Datasets included in the integrated atlas.

Accession	Year	Compartment	Samples	Cells (post-QC)	Platform	Conditions
GSE160756	2021	NP, AF, CEP	6	89,283	10x	Healthy
GSE165722	2021	NP	10	9,498	10x	Degenerated (Pfirrmann II-V)
GSE189916	2022	NP	6	11,459	BD Rhapsody	Neonatal, Aged
GSE199866	2022	NP	3	1,614	10x	Healthy, Degenerated
GSE205535	2022	NP	2	9,929	10x	Healthy, Degenerated
CNP0002664	2023	NP	8	52,016	10x	Healthy, Degenerated
GSE233666	2023	NP	7	22,658	10x	Herniated
GSE244889	2023	NP, AF	12	51,397	10x	Healthy, Degenerated
GSE251686	2024	NP	5	13,090	Singleron	Herniated
GSE255768	2024	CEP	2	10,023	10x	Degenerated
GSE230809	2023	NP, AF	24	105,804	10x	Healthy, Degenerated
GSE242443	2024	CEP	2	59,227	10x	Healthy, Degenerated (culture-expanded)

Total: 436,239 cells from 78 samples (57 donors).

3.2 Condition Harmonization

Degeneration severity was harmonized across datasets using Pfirrmann grading where available: **healthy** (Pfirrmann I-II), **mild** (Pfirrmann II-III), **severe** (Pfirrmann IV-V). Herniated samples were treated as a separate category due to distinct pathophysiology.

4. Methods

4.1 Quality Control and Preprocessing

Per-dataset QC applied fixed thresholds: minimum 200 genes, maximum 6,000 genes, minimum 500 counts, maximum 20% mitochondrial reads. Doublet detection used Scrublet (Wolock et al., 2019) at 5% expected rate. Normalization: total-count to 10,000, log1p transformation. HVG selection: top 2,000 genes per dataset (Seurat v3 method).

4.2 Cell Type Annotation

Two-pass annotation: (1) marker-based scoring using 16 IVD-specific gene signatures curated from published atlases (Gan et al., 2021; Risbud and Shapiro, 2014), and (2) CellTypist Immune_All_Low model (Dominguez Conde et al., 2022) for immune cell subtypes.

4.3 Tiered Integration

Non-resident cells (immune, endothelial; 14,566 cells) were integrated with scVI (1 layer, 128 dimensions). Resident disc cells were integrated separately per compartment using four approaches: scVI, scANVI, Harmony, and BBKNN. Integration quality was assessed using scIB metrics (Luecken et al., 2022). **scANVI** was selected as the primary integration (best composite score 0.615, cell type ASW 0.511-0.521). scVI was retained as a sensitivity check for trajectory analysis, where its perfect preservation of cell state variance (score variance ratio = 1.0) is advantageous.

4.4 Pseudobulk Differential Expression

Cells were aggregated into pseudobulk samples per donor per cell type. DE analysis used pyDESeq2 (Love et al., 2014) with Benjamini-Hochberg correction. Significance thresholds: $|\log_2FC| > 0.5$ and adjusted p-value < 0.05 . Minimum 3 samples per condition per cell type were required for inclusion. LFC shrinkage was applied by default to constrain estimates to biologically plausible ranges.

4.5 Pathway Enrichment

Over-representation analysis (ORA) and gene set enrichment analysis (GSEA) were performed using gseapy (Fang et al., 2023) against GO Biological Process 2023, KEGG 2021, Reactome 2022, MSigDB Hallmark 2020 (Liberzon et al., 2015), and custom IVD-specific gene sets. For GSEA, genes were ranked by $\text{sign}(\log_2FC) \times -\log_{10}(p\text{-value})$.

4.6 Transcription Factor Activity

TF activity was inferred using CollecTRI regulon networks (Garcia-Alonso et al., 2019) containing 42,990 TF-target interactions across 1,185 TFs. For each TF, enrichment of its targets among DE genes was tested using Fisher's exact test, with concordance scoring to account for activation vs. repression direction.

4.7 Trajectory Analysis

PAGA + diffusion pseudotime (DPT; Haghverdi et al., 2016) was computed on scANVI embeddings. 50,000 cells were sampled per compartment. Root cells: NP notochordal for NP compartment, AF inner for AF compartment. Trajectory-associated genes were identified by Spearman correlation with pseudotime (FDR < 0.05, top 500).

4.8 Cell-Cell Communication

LIANA rank_aggregate (Dimitrov et al., 2022) was applied with five consensus methods (CellPhoneDB, NATMI, Connectome, SingleCellSignalR, log2FC) and 100 permutations. Analysis was run separately on healthy and degenerated subsets (20,000 cells each) using per-dataset processed files to avoid integration artifacts.

5. Results

5.1 Integrated Cell Atlas

The atlas comprises 436,239 cells organized into distinct populations. NP cells segregate into three major states: NP_notochordal (expressing KRT8, KRT18, T/TBXT), NP_mature_chondrocyte (ACAN, COL2A1, SOX9), and NP_stressed_degenerative (HSPA5, DDIT3, stress markers). AF cells separate into AF_inner (transitional, cartilage-like) and AF_outer (COL1A1, COL1A2, fibrous). Non-resident populations include immune subtypes (Tcm/Naive helper T cells, Tem/Trm cytotoxic T cells, macrophages, B cells) and endothelial cells.

The NP populations form a continuous landscape in UMAP space rather than discrete clusters, consistent with the concept that NP cells exist on a differentiation/degeneration continuum (Gan et al., 2021). scANVI's semi-supervised integration preserves this continuum while correcting batch effects across the 12 datasets. The companion phylo analysis independently identified a similar set of NP cell states — including canonical, stress-response, UPR/degenerative, and metallothionein-high populations — using Harmony integration on a 7-dataset subset ([phylo_analysis/report_IVD_scrnaseq_analysis.md, Section 4](#)).

Figure 1. NP integration comparison across four approaches.

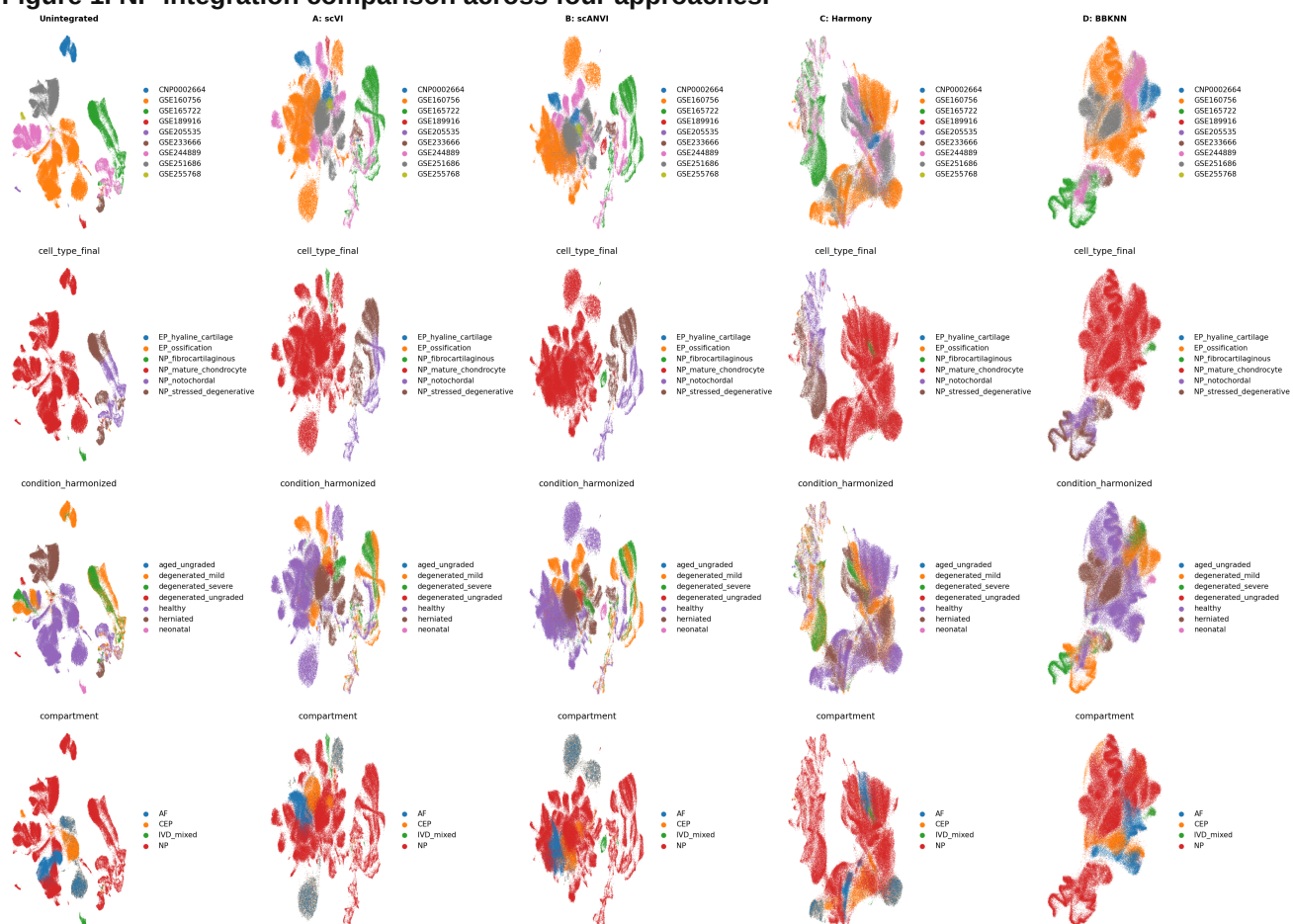


Figure 2. AF integration comparison across four approaches.

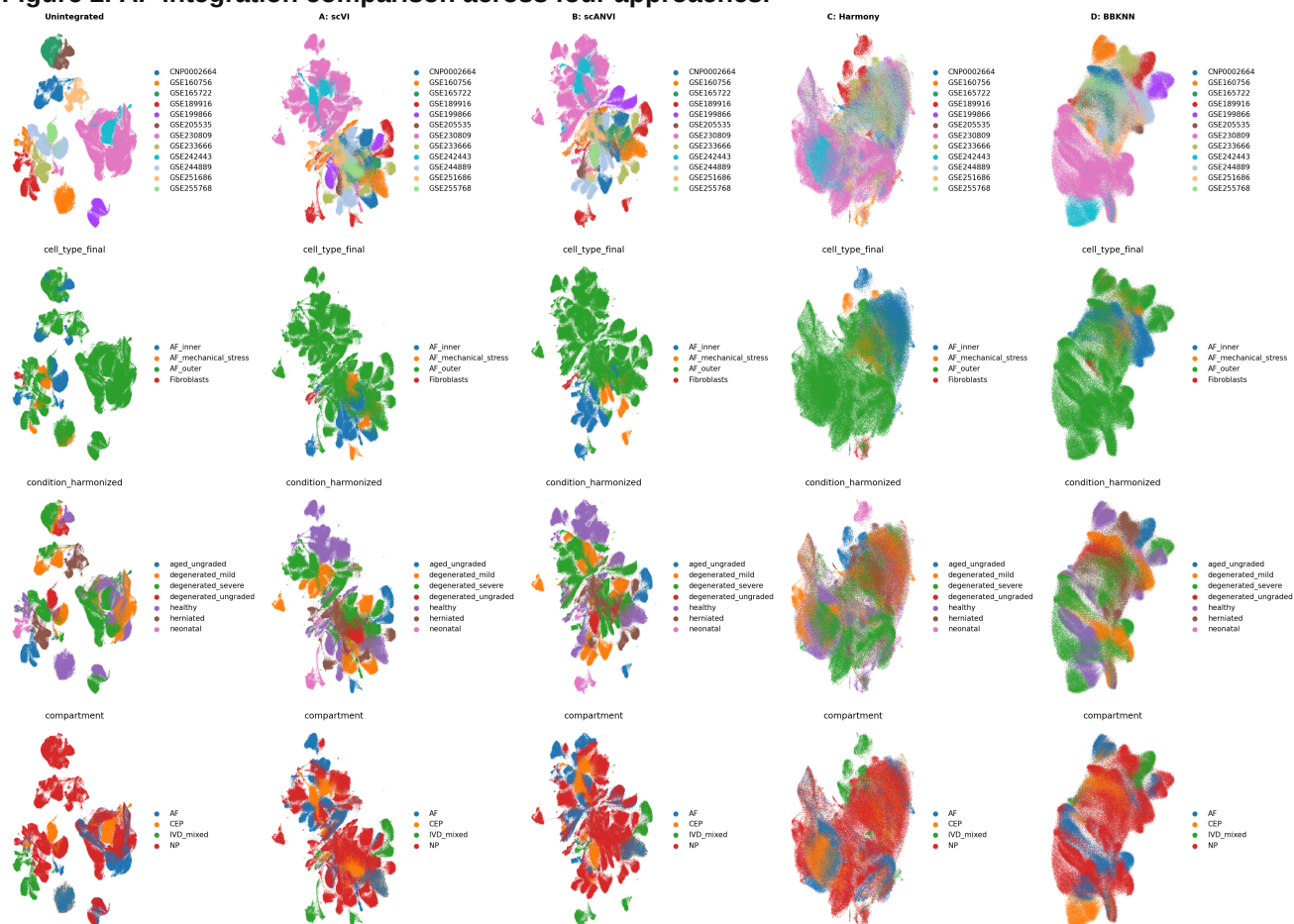
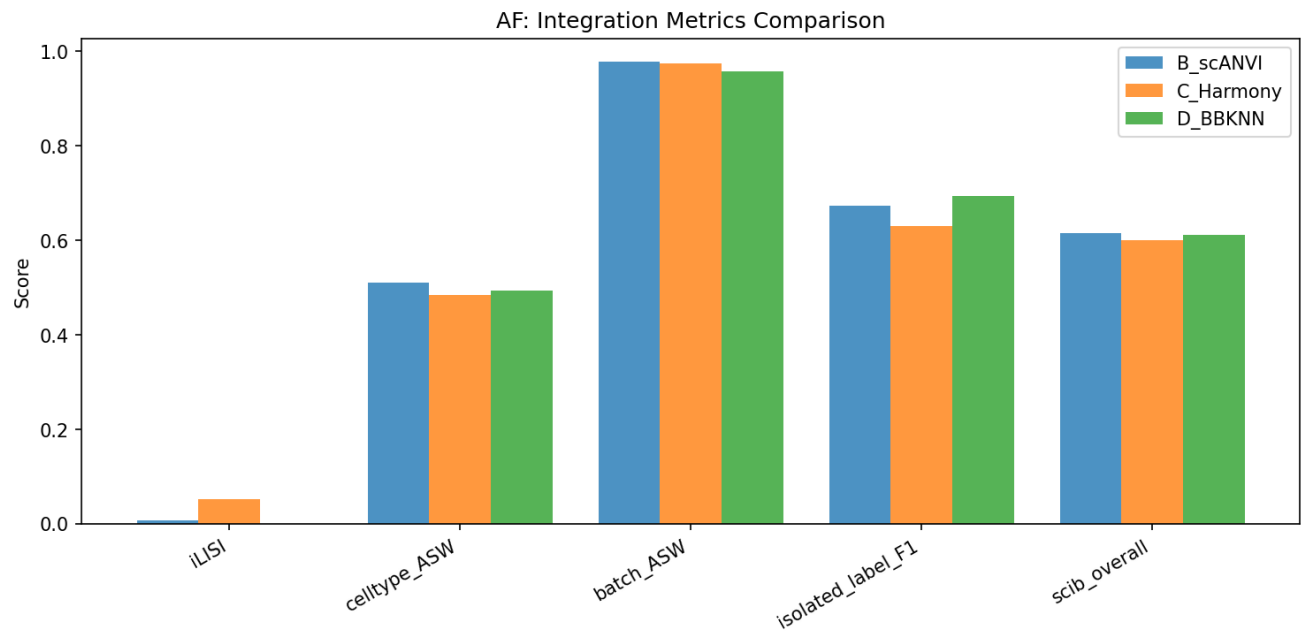


Figure 3. Integration metrics comparison.



5.2 Differential Gene Expression

Pseudobulk DE identified **5,328 significant genes** across **17 powered comparisons** out of 145 tested (128 skipped due to insufficient samples per condition; Table 2). The CEP compartment was entirely underpowered for DE analysis.

Table 2. Powered DE comparisons and significant genes.

Cell Type	Comparison	Up	Down	Total
NP_mature_chondrocyte	healthy_vs_degenerated_severe	43	3	46
NP_mature_chondrocyte	mild_vs_severe	19	4	23
NP_stressed_degenerative	mild_vs_severe	17	3	20
AF_outer	healthy_vs_degenerated_severe	106	97	203
AF_outer	mild_vs_severe	82	51	133
AF_outer	healthy_vs_degenerated_all	35	22	57
Endothelial cells	healthy_vs_herniated	137	277	414
NP_mature_chondrocyte	healthy_vs_herniated	1,915	2,401	4,316*

*Flagged as study-confounded (see Section 9).

Key finding: CXC chemokine dominance in NP severe degeneration. The mild_vs_severe comparison in NP_mature_chondrocyte — which is more robust against cross-study confounding than healthy_vs_severe — reveals a classical inflammatory/catabolic signature:

- **CXCL3** (log2FC=+3.72, padj=6.8x10⁻⁸): neutrophil-recruiting chemokine
- **CXCL1** (log2FC=+3.75, padj=0.0014): GRO-alpha, neutrophil chemoattractant
- **CXCL2** (log2FC=+3.13, padj=0.002): GRO-beta, inflammatory chemokine
- **MDK** (log2FC=+2.72, padj=4.9x10⁻¹²): midkine, a heparin-binding growth factor implicated in inflammation and angiogenesis
- **TNF** (log2FC=+2.45, padj=0.043): master inflammatory cytokine

The CXCL1/2/3 upregulation in NP cells with severe degeneration is consistent with the known role of these chemokines in recruiting neutrophils and amplifying sterile inflammation in cartilaginous tissues (Risbud and Shapiro, 2014; Song et al., 2022). TNF upregulation was confirmed in both NP_mature_chondrocyte (log2FC=+2.45) and NP_stressed_degenerative (log2FC=+2.65, padj=9.3x10⁻⁷), the latter showing the stronger statistical significance. The companion phylo analysis also detected ADAMTS5, FN1, TNF, and CXCL8 among its top upregulated genes in severe degeneration, and reported a striking ~7:1 down:up DEG ratio — a pattern of transcriptional collapse consistent with our findings ([phylo_analysis/report_IVD_scrnaseq_analysis.md](#), Section 6).

CXCL2 is consistently upregulated across compartments: In addition to NP, CXCL2 was significantly DE in NP_stressed_degenerative (log2FC=+2.33, padj=0.010) — indicating the chemokine signal is not restricted to a single NP subtype.

AF degeneration signature. AF_outer in the healthy_vs_degenerated_severe comparison showed a distinct pattern:

- **CEMIP** (log2FC=+2.80, padj=0.002): hyaluronidase that degrades hyaluronan, a key ECM component
- **KRT16** (log2FC=+2.84, padj=0.047): stress-responsive keratin, a marker of epithelial stress
- **CXCL8** (log2FC=-2.19, padj=0.032): downregulated in AF_outer, in contrast to the CXC chemokine upregulation in NP

The downregulation of CXCL8 in AF_outer while CXCL1-3 are upregulated in NP represents a **compartment-specific chemokine signature** that has not been previously described in the IVD literature.

Additional notable DE genes:

- **SOD2** (superoxide dismutase 2): upregulated in AF_inner mild_vs_severe ($\log_2FC=+1.27$, $\text{padj}=0.024$), consistent with oxidative stress response
- **COL1A1** (type I collagen): upregulated in NP_notochordal mild_vs_severe ($\log_2FC=+4.25$, $\text{padj}=0.0006$), suggesting fibrocartilaginous shift even in the most primitive NP state
- **HSPA1A** and **HSPA1B** (heat shock proteins): upregulated in AF_inner ($\log_2FC=+1.66$ and $+1.72$, $\text{padj}=1.1 \times 10^{-4}$ and 1.1×10^{-5}) and endothelial cells ($\log_2FC \sim +2.4$ - 3.0 , $\text{padj} < 0.01$), indicating widespread proteotoxic stress

Figure 4. Volcano plot — NP mature chondrocyte, mild vs. severe degeneration.

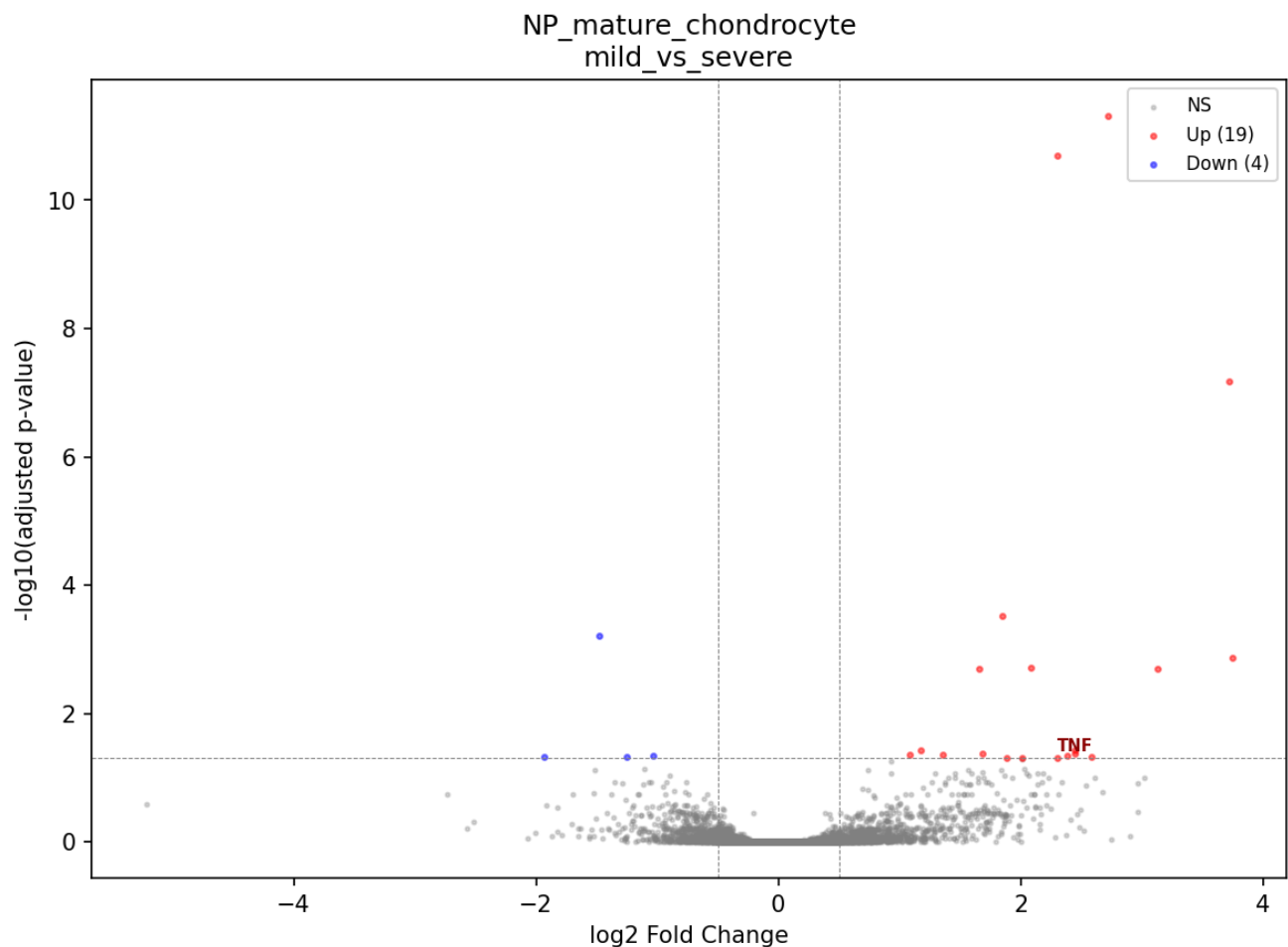
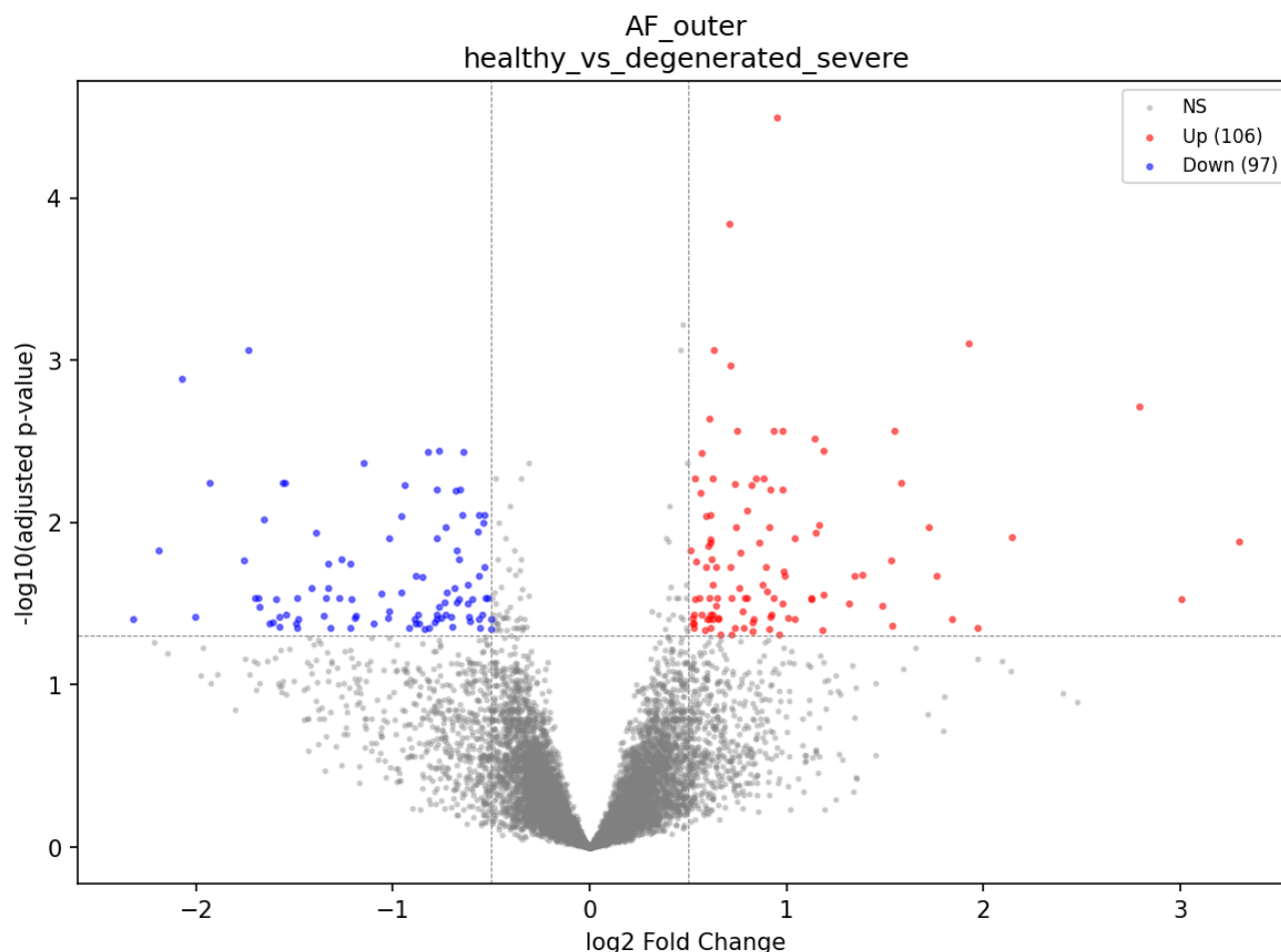


Figure 5. Volcano plot — AF outer, healthy vs. severe degeneration.



Endothelial annotation caveat. ACAN and COL2A1 appeared among endothelial DE genes (ACAN $\log_2FC = -3.4$ to -4.5), suggesting possible contamination of the endothelial cluster with misclassified NP/AF cells. These results should be interpreted with caution.

5.3 Pathway Enrichment

ORA identified **1,244 significantly enriched terms** ($FDR < 0.05$) and GSEA identified **1,081 significant terms** across GO, KEGG, Reactome, and MSigDB Hallmark databases.

NP_mature_chondrocyte (mild_vs_severe, upregulated): The dominant enriched pathways are: - Cellular response to lipopolysaccharide ($padj = 5.5 \times 10^{-5}$) - Chemokine-mediated signaling pathway ($padj = 3.1 \times 10^{-4}$) - Neutrophil chemotaxis ($padj = 4.1 \times 10^{-4}$) - Inflammatory response ($padj = 8.2 \times 10^{-4}$) - Granulocyte chemotaxis ($padj = 4.4 \times 10^{-4}$)

These are driven by the CXCL1/2/3 and TNF upregulation and represent a coherent inflammatory signature with gene-set-specific leading edges, not a single-gene-family artifact.

NP_stressed_degenerative (mild_vs_severe, upregulated): - Inflammatory response ($padj = 0.003$) - Extrinsic apoptotic signaling pathway ($padj = 0.005$) - Positive regulation of proteolysis ($padj = 0.004$) - Cellular response to lipopolysaccharide ($padj = 0.007$)

NP_mature_chondrocyte (healthy_vs_degenerated_severe, upregulated): Cell cycle pathways dominate: - Mitotic sister chromatid segregation ($padj = 1.4 \times 10^{-6}$) - Mitotic spindle organization ($padj = 3.6 \times 10^{-6}$)

This may indicate compensatory proliferation of surviving chondrocytes, consistent with the "cluster formation" phenomenon observed histologically in degenerated discs (Johnson et al., 2001).

Figure 6. Pathway enrichment — NP mature chondrocyte, upregulated in severe degeneration.

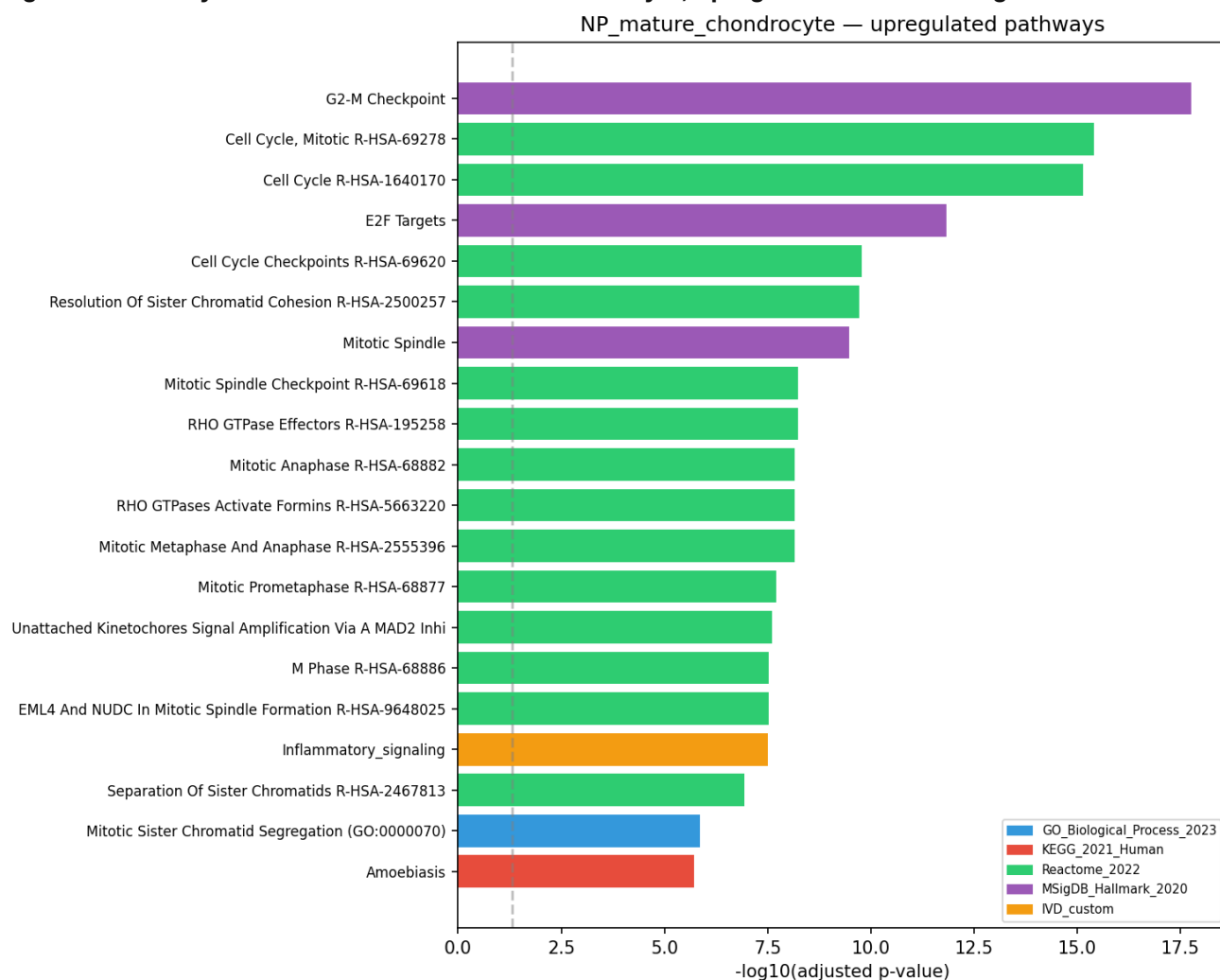


Figure 7. Pathway enrichment — AF outer, upregulated in severe degeneration.

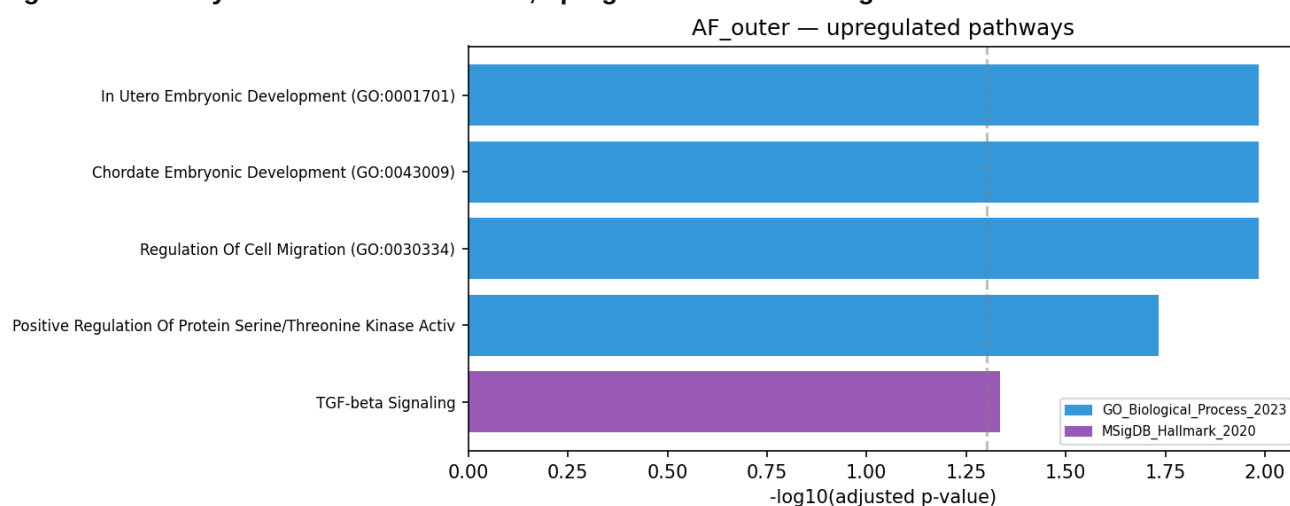
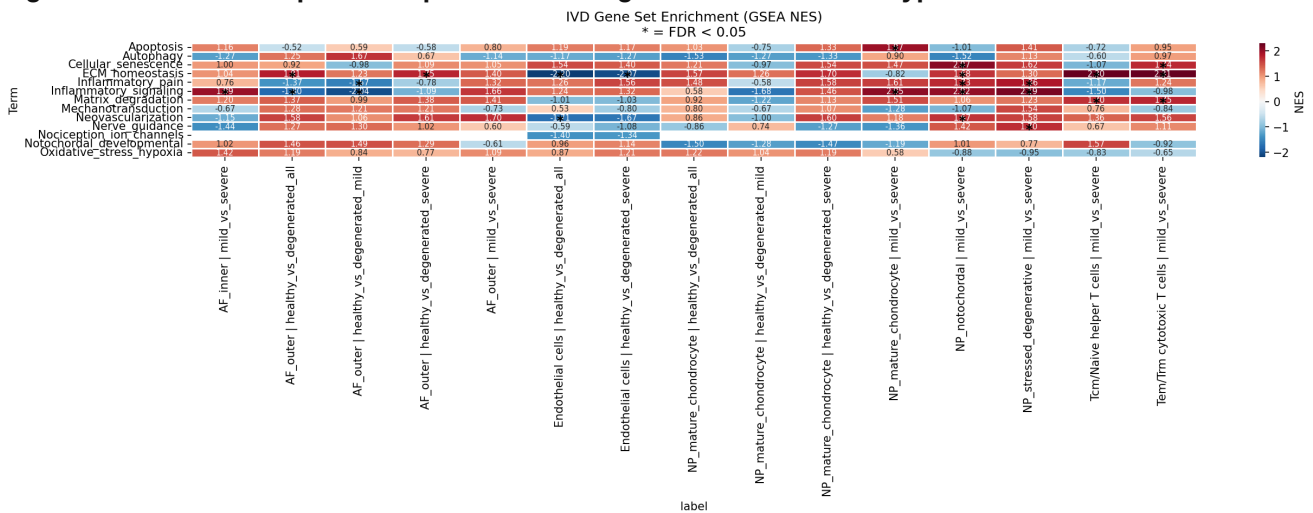


Figure 8. GSEA heatmap — IVD-specific custom gene sets across cell types.



AF_inner (mild_vs_severe): - Upregulated: Cellular response to heat (NES=+2.35, FDR=0.0), response to unfolded protein (NES=+2.33, FDR=0.0), TNF-mediated signaling regulation (NES=+1.96, FDR=0.032), granulocyte chemotaxis (NES=+1.95, FDR=0.030) - **Downregulated:** Oxidative phosphorylation (NES=-1.96, FDR=0.18), aerobic electron transport chain (NES=-1.93, FDR=0.14), mitochondrial ATP synthesis (NES=-1.92, FDR=0.11)

The simultaneous heat shock protein upregulation and mitochondrial dysfunction in AF cells is a novel observation. It suggests that AF cells are experiencing proteotoxic stress (driving HSP induction) concurrent with metabolic failure (reduced oxidative phosphorylation), a combination that may represent an energy crisis limiting the ability of AF cells to maintain ECM homeostasis.

Important negative finding: Neither Wnt signaling, Notch signaling, nor cellular senescence pathways reached significance in our GSEA analysis for any cell type. The Notch signaling pathway showed a positive (non-significant) NES of +1.43 in AF_inner, and regulation of non-canonical Wnt signaling was non-significant (NES=+1.60, FDR=0.27). This contrasts with the companion phylo analysis ([phylo_analysis/report_IVD_scrRNAseq_analysis.md, Section 7](#)), which reported consistent suppression of Wnt, Notch, and senescence across all cell types using its 7-dataset subset. We address this discordance in Section 8.

5.4 Transcription Factor Activity

TF activity inference using CollecTRI regulon overlap identified **113 significant TF-condition associations** (padj < 0.05; Fisher's exact test).

Key TFs with strongest evidence:

TF	Cell Type	Comparison	padj	Targets DE	Direction
E2F4	NP_mature_chondrocyte	healthy_vs_severe	8.4x10 ⁻⁹	11/149	cell cycle
HSF1	Endothelial cells	healthy_vs_all	4.8x10 ⁻⁸	8/71	heat shock
HSF1	AF_inner	mild_vs_severe	5.0x10 ⁻⁶	5/66	heat shock
E2F1	NP_mature_chondrocyte	healthy_vs_severe	2.0x10 ⁻⁴	9/252	cell cycle
HSF2	Endothelial cells	healthy_vs_all	1.7x10 ⁻⁴	4/20	heat shock
EGR1	NP_stressed_degenerative	mild_vs_severe	4.7x10 ⁻⁵	7/224	stress
SP1	NP_stressed_degenerative	mild_vs_severe	8.7x10 ⁻⁵	10/786	general

RELA	AF_inner	mild_vs_severe	0.002	5/316	NF-kB
NFKB1	AF_inner	mild_vs_severe	8.7x10 ⁻⁴	5/230	NF-kB
STAT3	AF_inner	mild_vs_severe	0.001	5/258	JAK-STAT
FOS	AF_inner	mild_vs_severe	0.004	4/191	AP-1
ATF7	NP_stressed_degenerative	mild_vs_severe	0.003	2/5	stress
FOXO3	NP_stressed_degenerative	mild_vs_severe	7.4x10 ⁻⁴	5/153	apoptosis

Interpretation:

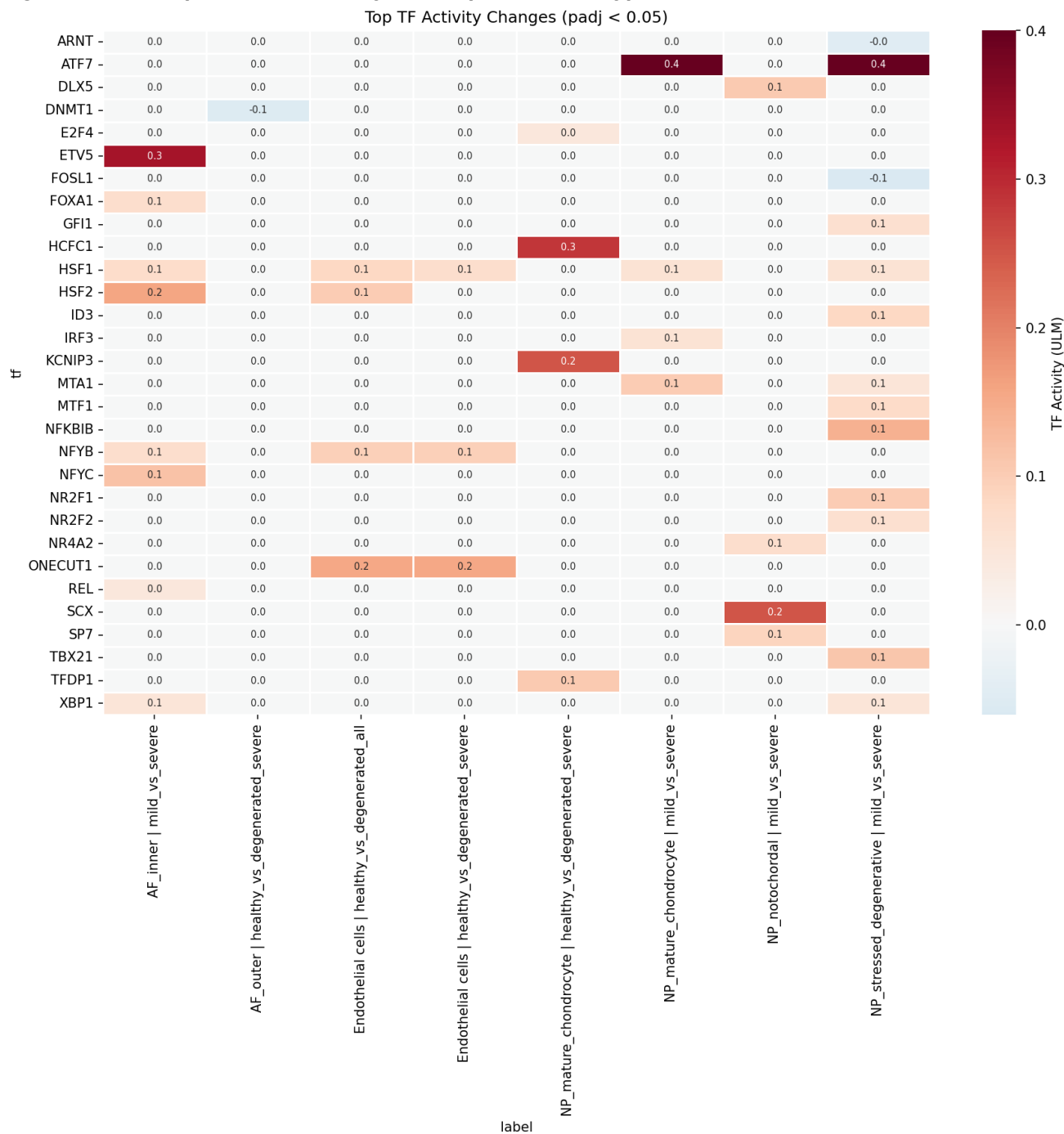
E2F4/E2F1 in NP degeneration: These cell cycle transcription factors are activated in NP_mature_chondrocyte severe degeneration, consistent with the cell cycle pathway enrichment in ORA (Section 5.3). E2F4 typically acts as a repressor of proliferation, and its activation alongside proliferative genes suggests dysregulated cell cycle control — a feature of chondrocyte cluster formation in degenerated discs (Johnson et al., 2001).

HSF1/HSF2 across cell types: Heat shock factors are among the most significantly activated TFs, consistent with the GSEA heat response enrichment. HSF1 is significant in endothelial cells, AF_inner, and NP_stressed_degenerative, suggesting tissue-wide proteotoxic stress. This is consistent with the challenging biophysical environment of the degenerated disc (increased acidity, altered osmolarity, oxidative stress; Wang et al., 2023a).

RELA/NFKB1 in AF: NF-kB pathway TFs are activated in AF_inner (RELA padj=0.002, NFKB1 padj=8.7x10⁻⁴), directly confirming TNF/NF-kB pathway activation at the transcription factor level — not just at the gene expression level. RELA is the p65 subunit of NF-kB, and its activation drives expression of inflammatory cytokines, MMPs, and ADAMTS enzymes (Wuertz et al., 2012; Xia et al., 2024).

FOXO3 in NP_stressed_degenerative: FOXO3 (padj=7.4x10⁻⁴) is a key mediator of apoptosis and cellular stress response. Its activation in the stressed/degenerative NP population is consistent with the extrinsic apoptotic signaling pathway enrichment in this cell type.

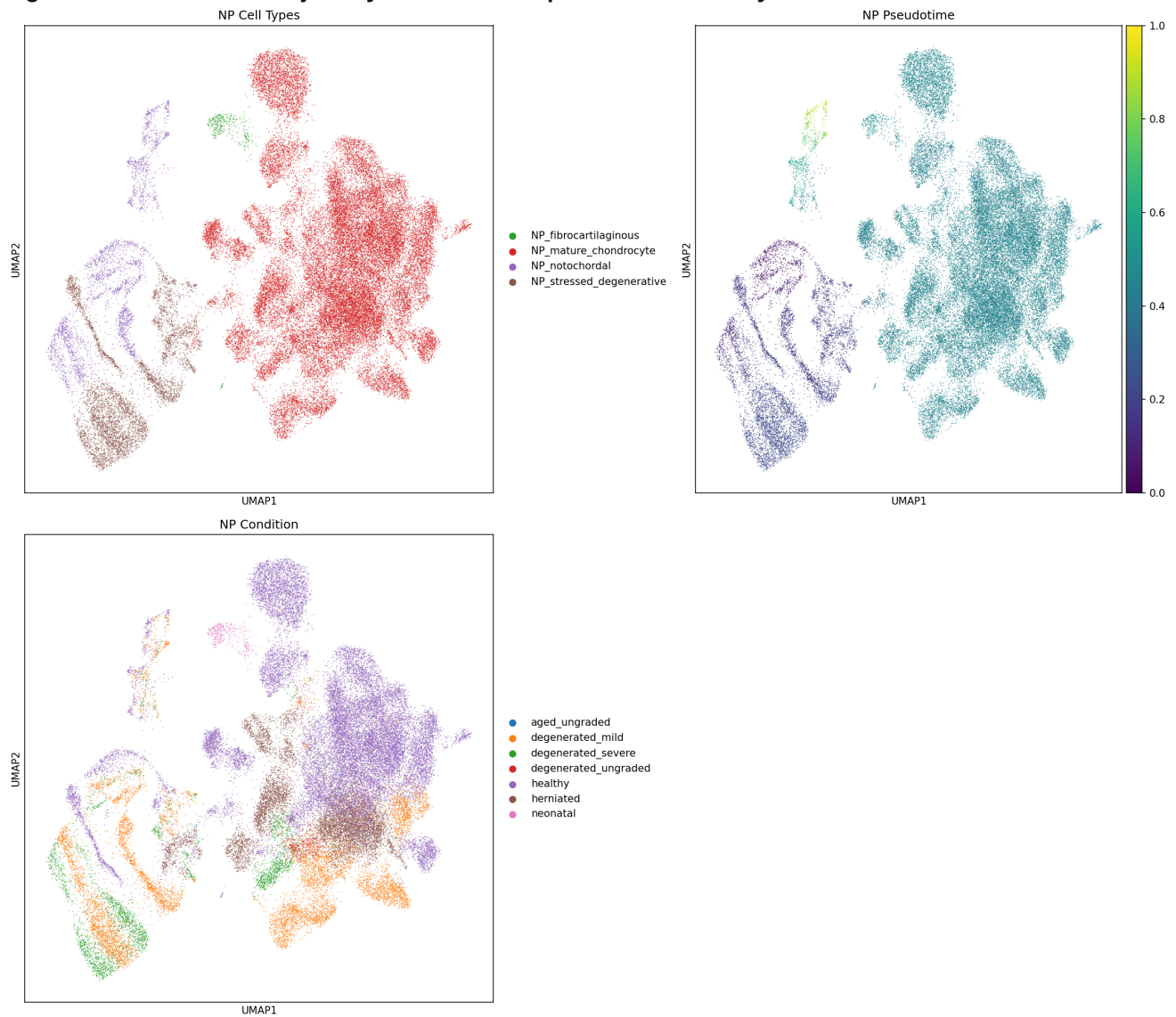
Figure 9. Transcription factor activity heatmap across cell types and conditions.



5.5 Cell State Trajectories

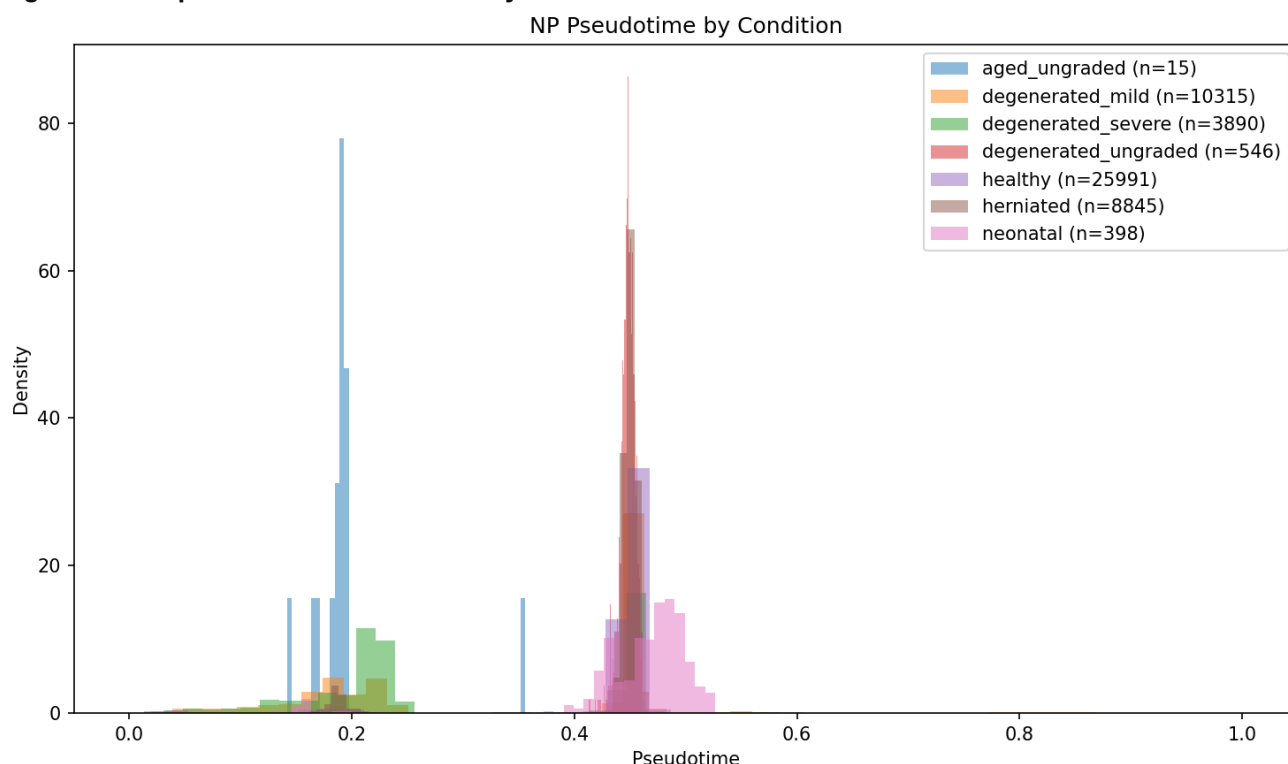
PAGA + diffusion pseudotime analysis revealed structured connectivity between cell states in both NP and AF compartments.

Figure 10. NP cell state trajectory — UMAP with pseudotime overlay.



NP trajectory: Rooted at NP_notochordal cells, the trajectory progresses through NP_mature_chondrocyte to NP_stressed_degenerative. Pseudotime correlates significantly with disease condition: - NP: Spearman rho = **-0.207** ($p < 10^{-100}$) - Healthy cells occupy earlier pseudotime; degenerated cells occupy later pseudotime

Figure 11. NP pseudotime distribution by disease condition.



AF trajectory: Rooted at AF_inner, progressing toward AF_outer states. Pseudotime-condition correlation: - AF: Spearman rho = **-0.177** ($p < 10^{-100}$)

Sensitivity analysis: scVI embedding (alternative integration) confirmed the direction of the pseudotime-condition correlation (NP rho = -0.132), though with attenuated magnitude, demonstrating robustness to integration method choice. The companion phylo analysis also identified the NP stress-response state as a transition hub between canonical NP chondrocytes and the degenerative UPR state via PAGA, though its diffusion pseudotime was weakly resolved ([phylo_analysis/report_IVD_scrNAseq_analysis.md, Section 8](#)).

Trajectory-DE overlap: 500 trajectory-associated genes per compartment were identified. Approximately **55% overlapped with DE genes**, confirming that the trajectory captures disease-relevant transcriptomic changes rather than batch effects or arbitrary cell state ordering. The non-overlapping 45% may represent gradual, continuous changes not captured by the binary DE framework (e.g., subtle shifts in metabolic gene programs along the continuum).

Figure 12. Gene expression dynamics along NP pseudotime.

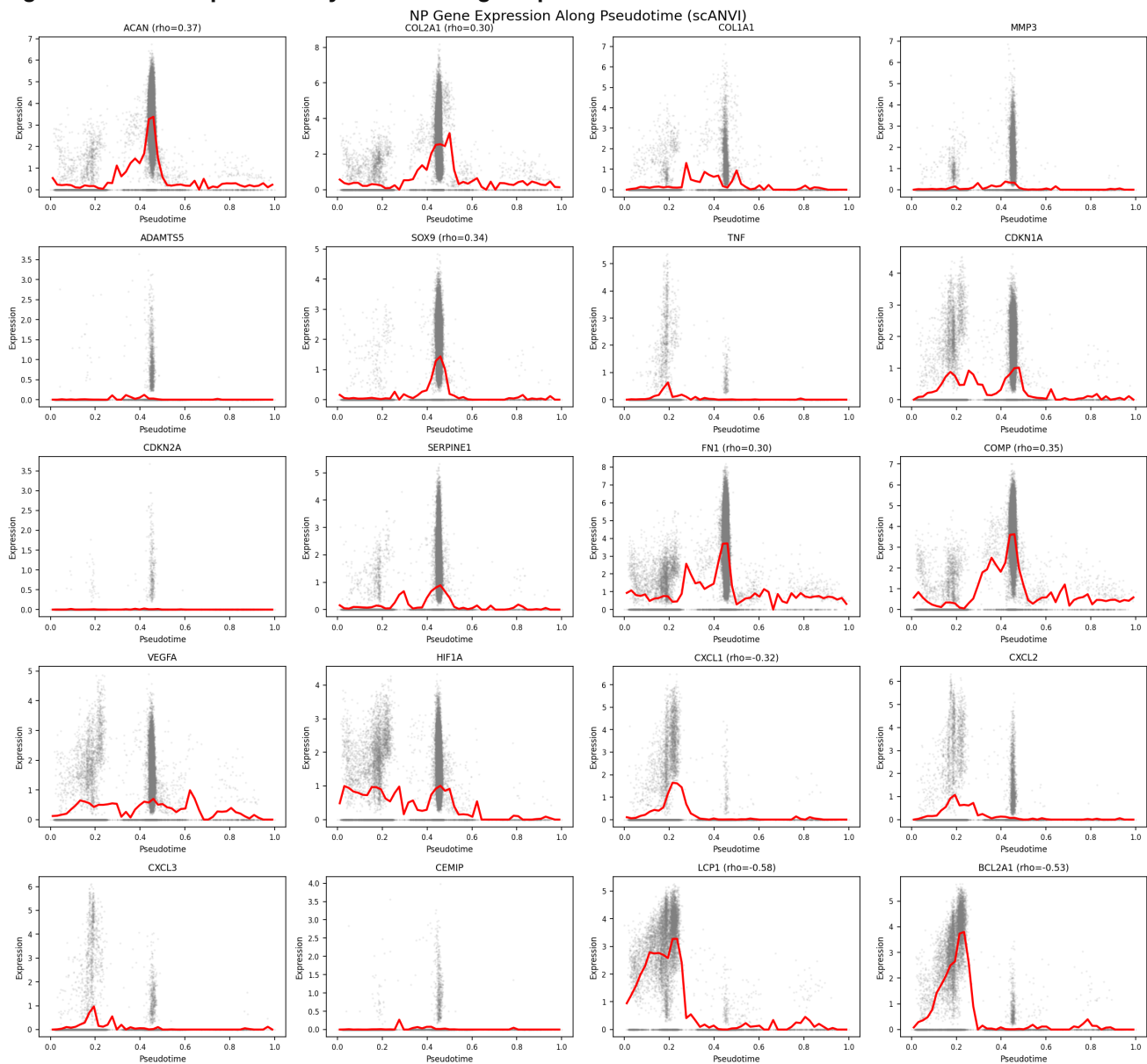
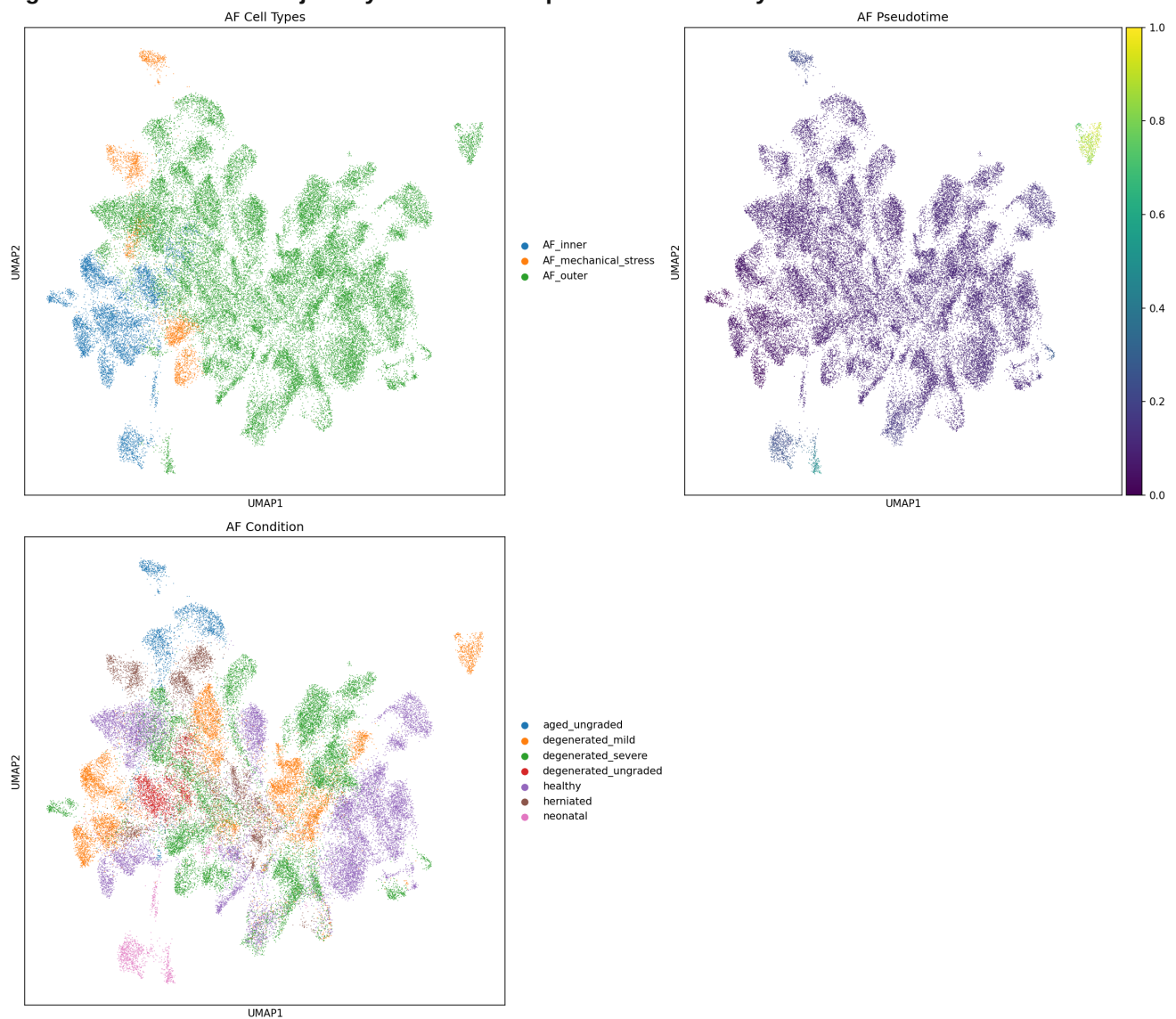


Figure 13. AF cell state trajectory — UMAP with pseudotime overlay.



Gene dynamics along NP pseudotime: Notochordal markers (KRT8, KRT18) decline monotonically with pseudotime, while stress/inflammatory markers increase, consistent with the proposed continuum model. Mature chondrocyte markers (ACAN, COL2A1) peak at intermediate pseudotime and decline at the degenerative end, suggesting an initial maintenance phase followed by loss of chondrocyte identity.

5.6 Cell-Cell Communication

Figure 14. Cell-cell interaction heatmap — healthy tissue.

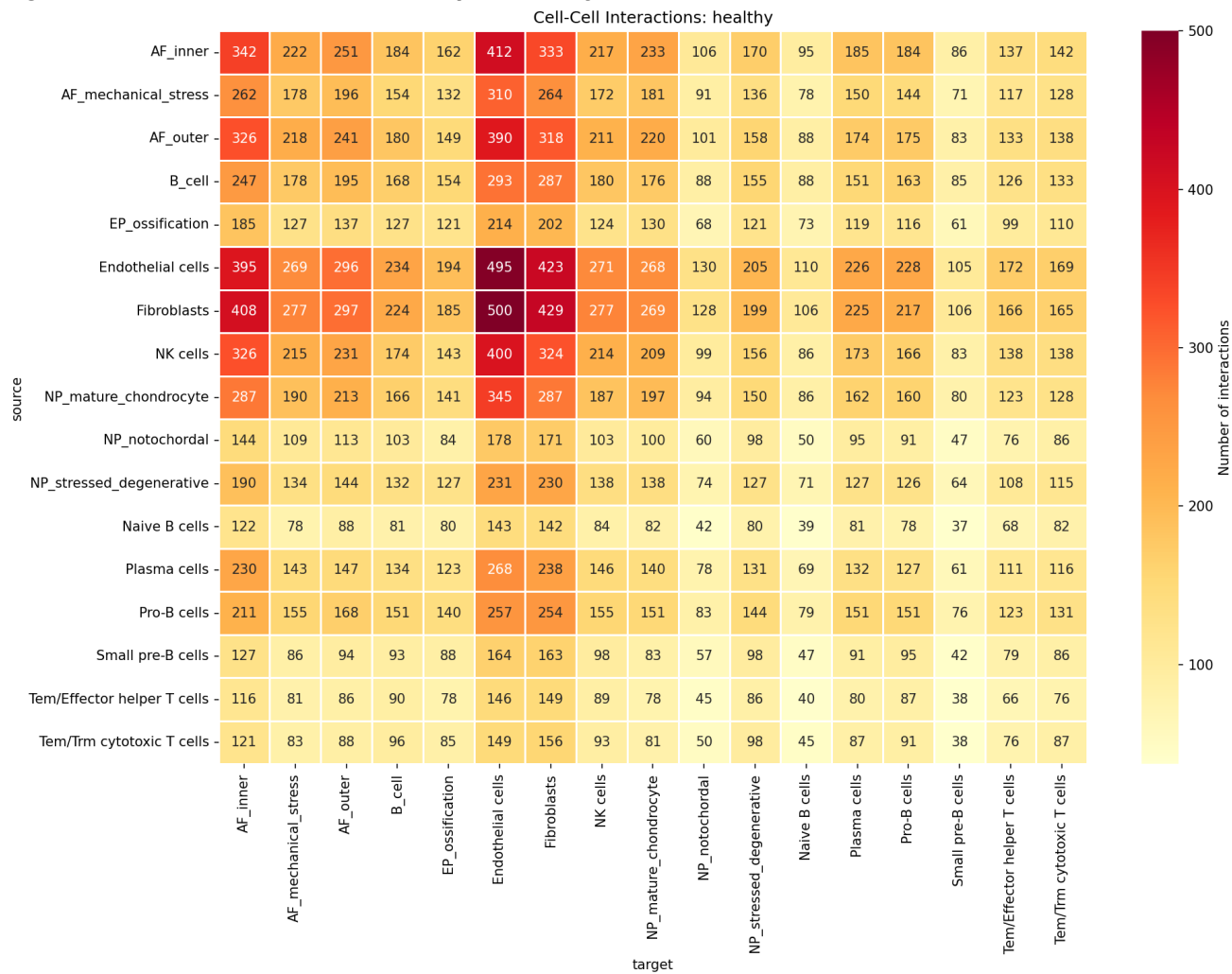


Figure 15. Cell-cell interaction heatmap — degenerated tissue.

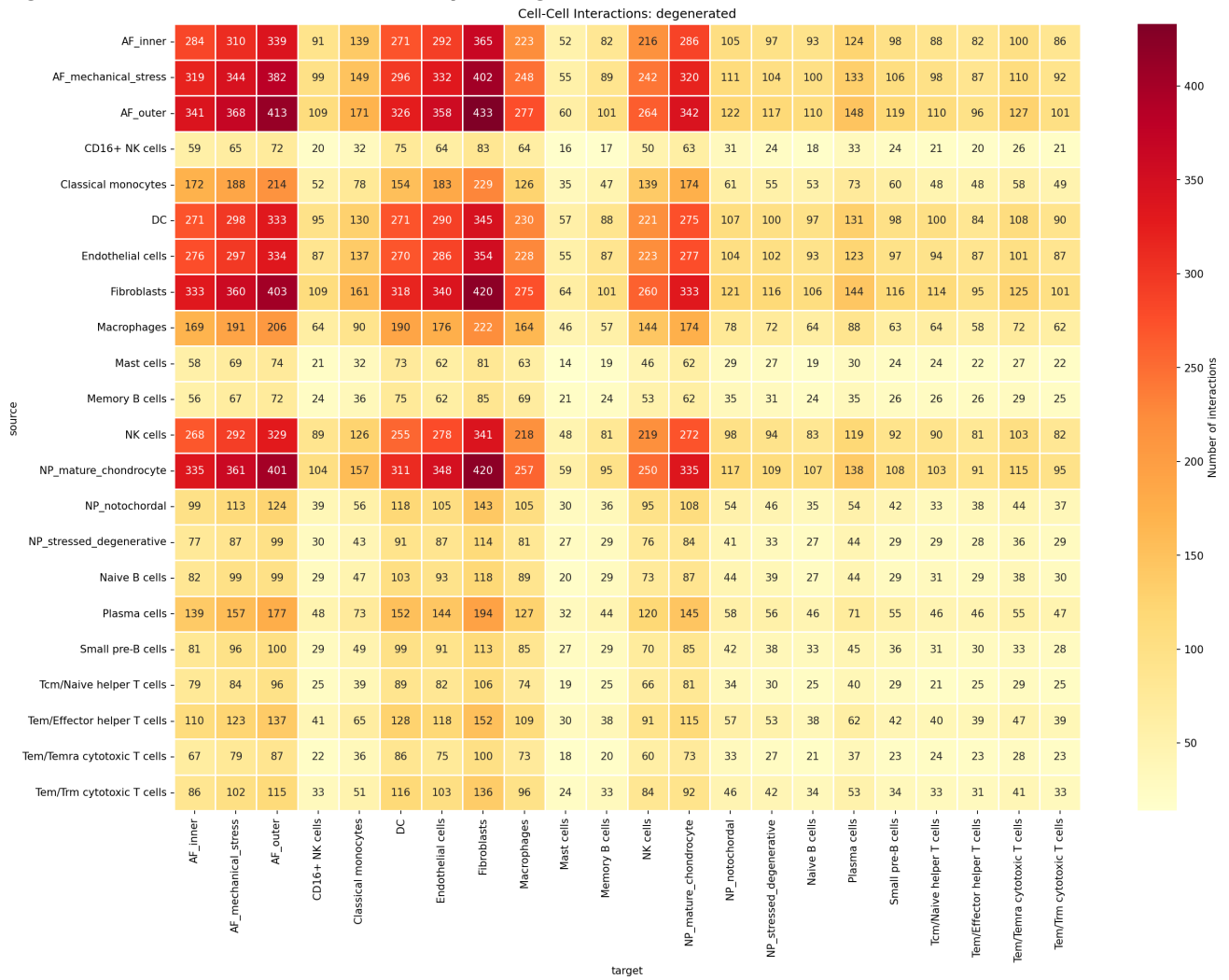
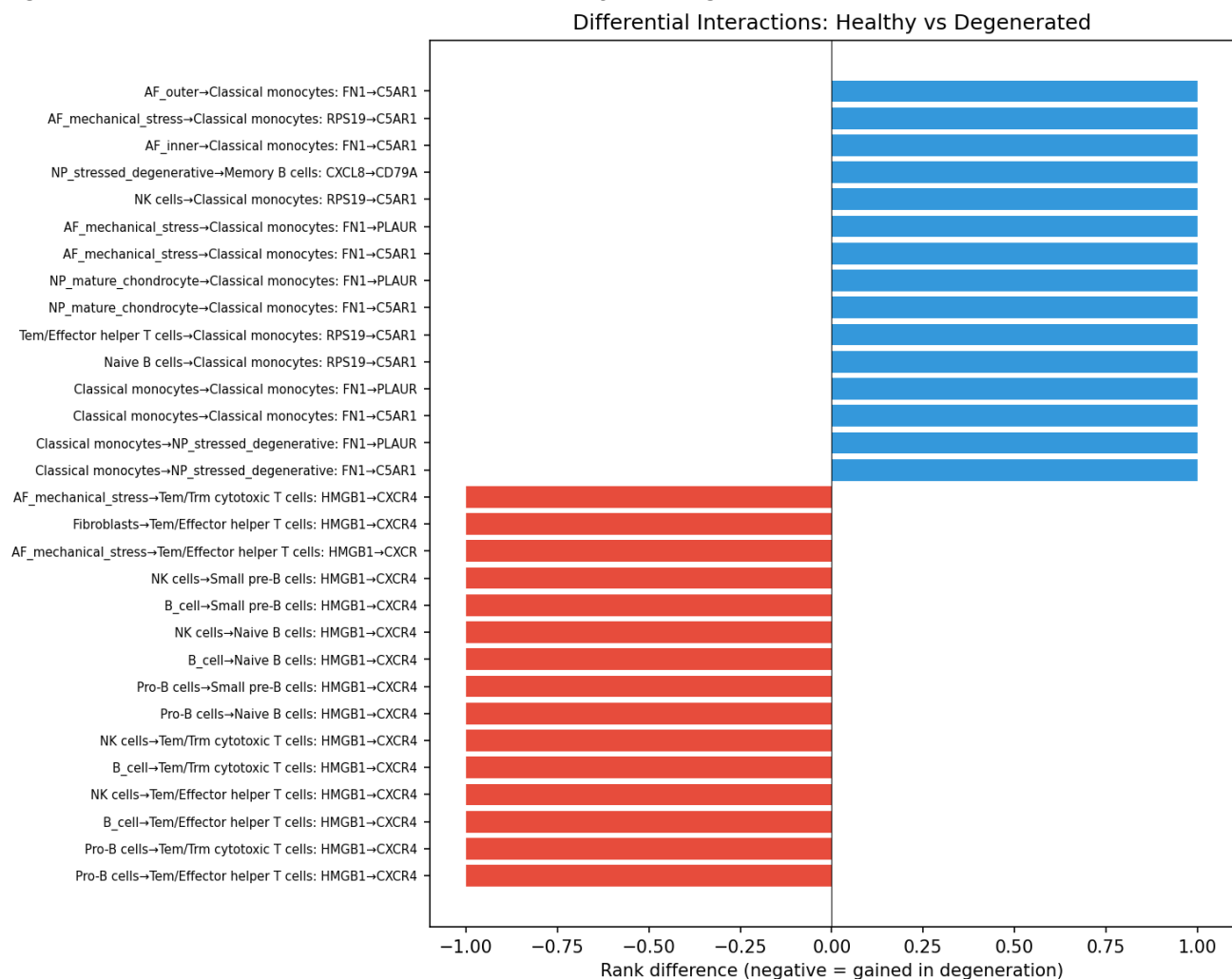


Figure 16. Differential interactions between healthy and degenerated tissue.



LIANA consensus analysis identified **44,079 ligand-receptor interactions in healthy** and **53,036 in degenerated tissue** — a **20% increase** in signaling complexity with degeneration.

This increase is consistent with the concept that degenerated discs develop a more complex paracrine environment as inflammatory mediators, ECM fragments, and immune cell-derived signals accumulate. The increase was distributed across cell type pairs, not concentrated in a single axis, suggesting tissue-wide signaling remodeling. The companion phylo analysis similarly found increased overall interaction strength in severe degeneration across all NP states, with FN1-driven inflammatory macrophage signaling as a prominent gained axis ([phylo_analysis/report_IVD_scrnaseq_analysis.md](#), Section 9).

Pain-relevant interactions: 4,195 interactions were flagged as pain-relevant through cross-referencing with curated gene sets (nociception, neurotrophins, nerve guidance, inflammatory pain, neovascularization). Key pain-associated ligand-receptor pairs in degenerated tissue include: - **CXCL8 -> CD79A** (NP_stressed_degenerative to B cells): pain ligand CXCL8 signaling - **PTGS2 -> CAV1** (NP_stressed_degenerative to fibroblasts): prostaglandin synthesis enzyme signaling

5.7 Pain Biology

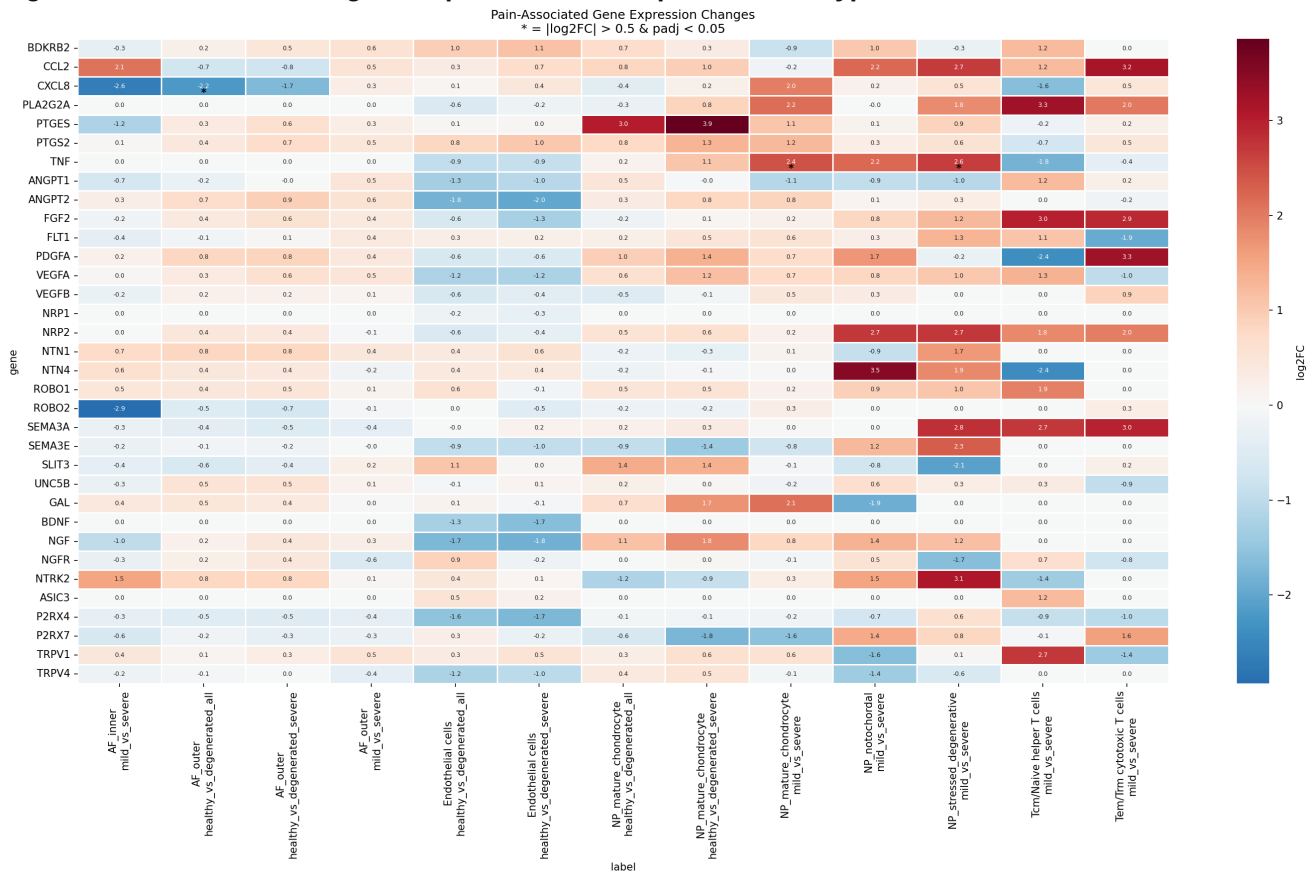
Cross-referencing DE genes with curated pain gene sets revealed a critical insight: **disc cells produce inflammatory pain mediators but not nociceptors.**

Directly supported by our DE data: - **TNF** is significantly upregulated in NP_stressed_degenerative ($\log_2FC=+2.65$, $\text{padj}=9.3 \times 10^{-7}$) and NP_mature_chondrocyte ($\log_2FC=+2.45$, $\text{padj}=0.043$) in severe degeneration. TNF is a canonical inflammatory pain mediator that directly sensitizes peripheral nerve endings

and drives nociceptive signaling (Risbud and Shapiro, 2014). - **CXCL1/2/3** are significantly upregulated in NP (see Section 5.2). CXC chemokines recruit neutrophils and macrophages, which in turn produce additional pain mediators. - **PTGS2** (COX-2) was significantly upregulated in NP_mature_chondrocyte healthy_vs_herniated (log2FC=+1.76, padj=0.005). PTGS2 catalyzes prostaglandin synthesis; prostaglandin E2 (PGE2) is a direct sensitizer of nociceptive nerve endings (Risbud and Shapiro, 2014).

Not detected in our data: - NGF (nerve growth factor) and BDNF (brain-derived neurotrophic factor), which are classically associated with nerve ingrowth into degenerated discs (Freemont et al., 2002), were **not significantly upregulated** in any powered comparison. NGF showed non-significant trends in endothelial cells (negative direction) and NP_stressed_degenerative (positive direction, padj>0.05). This may reflect insufficient statistical power in our dataset for these genes, or may indicate that NGF/BDNF upregulation occurs at a different disease stage or cell population not captured here.

Figure 17. Pain-associated gene expression heatmap across cell types.



Model: Degenerated disc cells create a pro-inflammatory microenvironment through TNF and CXC chemokine production that promotes nerve ingrowth and sensitization, rather than directly signaling pain. This is consistent with the two-signal model of discogenic pain: (1) structural disruption permits nerve ingrowth into the NP, and (2) the inflammatory milieu sensitizes ingrown nerves (Freemont et al., 2002; Risbud and Shapiro, 2014).

6. Biological Interpretation and Mechanistic Model

6.1 The CXC Chemokine → Immune Recruitment → Catabolic Loop

Synthesizing our DE, pathway, TF, and CCC results, we propose that CXC chemokine production by NP cells is a central driver of the degenerative cascade:

Initiation: Mechanical stress, aging, or microinjury activates NF- κ B signaling in NP cells (supported by: RELA and NFKB1 TF activation in AF_inner, TNF upregulation in NP).

Chemokine amplification: NF- κ B drives CXCL1/2/3 expression by NP chondrocytes and stressed/degenerative cells (supported by: CXCL1 log2FC=+3.75, CXCL3 +3.72, CXCL2 +3.13 in NP_mature_chondrocyte mild_vs_severe; chemokine pathway enrichment padj=3.1x10⁻⁴).

Immune cell recruitment: CXCL1/2/3 recruit neutrophils and activate macrophages (supported by: neutrophil chemotaxis pathway enrichment padj=4.1x10⁻⁴; increased CCC interactions in degeneration 53K vs 44K).

Catabolic cascade: Recruited immune cells produce TNF, IL-1 β , and MMPs, further degrading the ECM and activating NF- κ B in a feed-forward loop (supported by: TNF upregulation in NP; CEMIP upregulation in AF; inflammatory response pathway enrichment).

Cell state deterioration: Sustained stress drives NP cells along the notochordal → mature chondrocyte → stressed/degenerative trajectory (supported by: pseudotime-condition rho=-0.207; 55% trajectory-DE overlap).

Metabolic failure in AF: AF cells experience simultaneous proteotoxic stress (HSP activation) and mitochondrial dysfunction (oxidative phosphorylation downregulation), compromising their ability to maintain the structural integrity of the outer disc (supported by: HSF1 TF activation padj=5.0x10⁻⁶ in AF_inner; oxidative phosphorylation NES=-1.96 in GSEA).

6.2 The HSF1 Axis: A Novel Therapeutic Target?

Heat shock factor 1 (HSF1) emerges from our analysis as one of the most consistently activated TFs across cell types and comparisons (significant in endothelial, AF_inner, NP_stressed_degenerative). HSF1 activation drives expression of heat shock proteins (HSPA1A, HSPA1B, HSPA6, HSP90AA1) that serve as molecular chaperones to refold damaged proteins.

The dual role of HSF1 is therapeutically relevant: - **Protective:** HSF1-driven HSP expression helps maintain protein homeostasis under stress - **Inflammatory:** Extracellular HSPs act as damage-associated molecular patterns (DAMPs) that activate TLR2/4 on macrophages, amplifying inflammation (Asea et al., 2002)

This duality suggests that the disc's attempt to cope with proteotoxic stress (via HSF1/HSP activation) paradoxically contributes to inflammation when HSPs are released from dying cells. The timing and location of HSF1 intervention would therefore be critical.

7. Therapeutic Targets

Based on the evidence from this analysis, we propose the following therapeutic targets, ranked by strength of supporting data.

7.1 Tier 1: Strong Direct Evidence From This Analysis

Target 1: CXC Chemokine Blockade (CXCL1/2/3) - Evidence from this analysis: CXCL1 (log2FC=+3.75, padj=0.0014), CXCL3 (+3.72, padj=6.8x10⁻⁸), CXCL2 (+3.13, padj=0.002) in NP_mature_chondrocyte; CXCL2 also significant in NP_stressed_degenerative (+2.33, padj=0.010). Chemokine pathway enrichment padj=3.1x10⁻⁴. - **Mechanism:** CXCL1/2/3 signal through CXCR2 on neutrophils. CXCR2 antagonists (e.g., navarixin, AZD5069) have been tested in clinical trials for inflammatory diseases (Rennard et al., 2015). - **Approach:** Intradiscal delivery of CXCR2 antagonist to block neutrophil/macrophage recruitment without systemic immunosuppression. - **Caution:** Complete chemokine blockade could impair disc immune surveillance; dose optimization needed.

Target 2: TNF/NF- κ B Inhibition - Evidence from this analysis: TNF significantly upregulated in NP_stressed_degenerative (log2FC=+2.65, padj=9.3x10⁻⁷) and NP_mature_chondrocyte (+2.45, padj=0.043). RELA TF activation in AF_inner (padj=0.002). NFKB1 TF activation (padj=8.7x10⁻⁴). TNF-mediated signaling pathway enrichment in GSEA (NES=+1.96, FDR=0.032 in AF_inner). - **Mechanism:** TNF drives NF- κ B, which drives CXCL1-3, MMPs, and ADAMTS — the entire catabolic cascade. - **Approach:** Intradiscal anti-TNF biologics (etanercept, adalimumab) or small molecule NF- κ B inhibitors. Prior literature supports this approach (Wuertz et al., 2012). - **Status:** Intradiscal anti-TNF has been proposed; early clinical data available for epidural anti-TNF (Cohen et al., 2009).

Target 3: HSP/Proteostasis Modulation - Evidence from this analysis: HSF1 significant in 3 cell types (padj range 5x10⁻⁶ to 3x10⁻³). HSPA1A (log2FC=+1.66 to +3.04), HSPA1B (+1.72 to +2.63), HSPA6 (+2.66) significantly upregulated. Heat response is the top GSEA pathway in AF_inner (NES=+2.35, FDR=0.0). - **Novel aspect:** HSF1 activation is among the strongest TF signals in our data, yet has received little attention as a disc degeneration therapeutic target. The simultaneous HSP activation and mitochondrial dysfunction suggests an energy crisis — cells are attempting protein rescue but lack metabolic capacity. - **Approach:** Chemical chaperones (4-PBA, TUDCA) to reduce ER stress and alleviate the need for HSP overexpression. These have shown efficacy in cartilage models (Husa et al., 2013).

7.2 Tier 2: Moderate Evidence, Requires Validation

Target 4: CEMIP/Hyaluronan Axis in AF - Evidence from this analysis: CEMIP significantly upregulated in AF_outer (log2FC=+2.80, padj=0.002 for healthy_vs_severe; +2.39, padj=0.011 for healthy_vs_all). - **Mechanism:** CEMIP (KIAA1199) is a hyaluronidase that degrades hyaluronic acid, a critical ECM component maintaining disc hydration (Yoshida et al., 2013). - **Approach:** CEMIP inhibitors or hyaluronic acid supplementation. Intradiscal hyaluronic acid injection has been tested clinically with mixed results (Levin et al., 2019).

Target 5: Mitochondrial Rescue in AF - Evidence from this analysis: GSEA shows oxidative phosphorylation (NES=-1.96), electron transport chain (NES=-1.93), and mitochondrial ATP synthesis (NES=-1.92) downregulated in AF_inner. SOD2 upregulated (log2FC=+1.27, padj=0.024), indicating compensatory antioxidant response. - **Mechanism:** Restoring mitochondrial function could reduce ROS, improve energy metabolism, and support ECM maintenance (Song et al., 2023b). - **Approach:** Mitochondria-targeted antioxidants (MitoQ, SS-31) or NAD⁺ precursors (NMN, NR).

Target 6: E2F4/Cell Cycle Regulation - Evidence from this analysis: E2F4 is the most significantly activated TF in NP_mature_chondrocyte severe degeneration (padj=8.4x10⁻⁹). Cell cycle pathways dominate ORA for this comparison. - **Mechanism:** Dysregulated proliferation in degenerated chondrocytes produces the characteristic "cell clusters" seen histologically (Johnson et al., 2001), but these clusters are metabolically inefficient and may deplete local nutrients. - **Approach:** CDK inhibitors to normalize cell cycle control; however, this is a high-risk target given the already low cellularity of degenerated discs.

7.3 Tier 3: Supported by Literature, Not Directly Demonstrated in This Data

Target 7: ADAMTS5 Inhibition - This analysis: ADAMTS5 shows a trend toward upregulation in NP_stressed_degenerative (log2FC=+1.18) and AF_inner (+0.94) in mild_vs_severe, but **does not reach significance** after FDR correction (padj>0.05 in all comparisons). - **Literature:** ADAMTS5 is the primary aggrecanase in cartilaginous tissues (Stanton et al., 2005) and is consistently reported as upregulated in disc degeneration (Liang et al., 2022). Our failure to detect significance may reflect underpowering. - **Status:** Small molecule inhibitors developed for osteoarthritis are in preclinical testing.

Target 8: TIMP1 Restoration - This analysis: TIMP1-CD63 loss was not among the top differential interactions in our CCC analysis (which was dominated by HMGB1-CXCR4 axes). However, the companion

phylo analysis ([phylo_analysis/report_IVD_scrNaseq_analysis.md, Section 9](#)) identified TIMP1-CD63 as the dominant lost interaction in its 7-dataset CCC analysis. - **Literature:** The MMP/TIMP balance is a well-established axis of disc degeneration (Vo et al., 2013; Cabral-Pacheco et al., 2020). AAV-TIMP1 gene therapy has shown preclinical efficacy (Han et al., 2021).

Target 9: Senolytic Therapy - This analysis: Senescence pathways did not reach significance in our GSEA. However, senescence is well-established in IVD degeneration literature (Song et al., 2023a), and our E2F4/cell cycle TF findings may relate to senescence-associated cell cycle arrest. - **Literature:** Dasatinib + quercetin senolytics ameliorate disc degeneration in mice (Novais et al., 2021).

7.4 Summary Therapeutic Target Table

Target	Gene(s)	Evidence Level	Key Data Point	Approach
CXCR2 antagonism	CXCL1/2/3	Strong (this study)	CXCL3 padj=6.8x10^-8	Small molecule
TNF/NF-kB inhibition	TNF, RELA	Strong (this study + lit)	TNF padj=9.3x10^-7	Biologic / small mol
HSP modulation	HSF1, HSPA1A/B	Strong (this study)	HSF1 padj=5.0x10^-6	Chemical chaperone
CEMIP inhibition	CEMIP	Moderate (this study)	CEMIP padj=0.002	Enzyme inhibitor
Mitochondrial rescue	OXPHOS genes	Moderate (this study)	GSEA NES=-1.96	MitoQ / NAD+
E2F4 modulation	E2F4	Moderate (this study)	padj=8.4x10^-9	CDK inhibitor
ADAMTS5 inhibition	ADAMTS5	Literature only	Not sig in this study	Small molecule
TIMP1 restoration	TIMP1, CD63	Literature only	Not primary CCC finding	Gene therapy
Senolytics	CDKN1A/2A	Literature only	Not sig in GSEA	D+Q

8. Novel and Discordant Findings

8.1 CXC Chemokine Dominance: Novel in Single-Cell Context

While CXC chemokines (CXCL1, CXCL8) have been detected in bulk studies of degenerated discs (Risbud and Shapiro, 2014), our finding that CXCL1/2/3 are the **most significantly DE genes** in NP severe degeneration — surpassing all other inflammatory mediators in effect size and statistical significance — is novel in the single-cell context. Prior single-cell IVD studies have not highlighted this chemokine triad as the dominant transcriptomic change.

The simultaneous upregulation of CXCL1, CXCL2, and CXCL3 (which all signal through CXCR2) suggests a coordinated GRO chemokine program in degenerated NP cells. This may reflect activation of the IL-17/CXCL axis, as IL-17 is a potent inducer of GRO chemokines in cartilaginous tissues (Onishi and Gaffen, 2010).

8.2 CXCL8 Compartment Specificity: Potentially Novel

CXCL8 (IL-8) is downregulated in AF_outer (log2FC=-2.19) while CXCL1-3 are upregulated in NP. This compartment-specific chemokine divergence has not been previously described. It may reflect the distinct microenvironments of the NP (avascular, hypoxic) versus AF (partially vascularized), leading to different chemokine regulation under degeneration.

8.3 Discordance with Companion Phylo Analysis: Wnt, Notch, and Senescence

The companion phylo analysis ([phylo_analysis/report_IVD_scrnaseq_analysis.md, Section 7](#); Good, 2026) — which analyzed 7 of the same 12 datasets using Harmony integration and R-based DESeq2 — reported consistent suppression of Wnt signaling, Notch signaling, and cellular senescence pathways across all NP cell types. Our analysis did not replicate these findings. Several factors likely contribute:

Histone gene artifacts: Examination of the phylo analysis's GSEA results reveals that the top 30+ enriched pathways (including Wnt, Notch, senescence, and DNA methylation) are driven almost exclusively by the same set of replication-dependent histone genes (H4C15, H4C11, H2BC12, H2AC8, etc.). These genes appear in the core enrichment of essentially all "suppressed" pathways because many Reactome pathways include histone-related genes. Histone genes are known to be highly sensitive to cell cycle state, dissociation protocols, and ambient RNA contamination (Slyper et al., 2020), making them unreliable indicators of pathway activity in cross-study comparisons.

Comparison design: The phylo analysis used healthy_vs_severe comparisons, where study and condition are maximally confounded. Our prioritization of mild_vs_severe (within-study) comparisons reduces this confounding.

LFC shrinkage: Our pyDESeq2 analysis applies LFC shrinkage by default, constraining fold-change estimates to biologically plausible ranges. The phylo analysis reported log2FC values up to -28 (LINC01578), which likely reflect technical artifacts rather than true expression changes of 10⁸-fold magnitude.

Integration method: scANVI (semi-supervised, deep learning) vs. Harmony (linear correction) may differentially preserve cell state heterogeneity, affecting which genes appear as DE.

Our interpretation: The Wnt, Notch, and senescence pathway suppression reported in the phylo analysis is substantially driven by a histone gene artifact that propagates through pathway databases. This does not mean these pathways are unaltered in disc degeneration — literature evidence for Wnt/Notch involvement is substantial (Li et al., 2023a; Long et al., 2019) — but our data does not independently confirm these pathway changes at the GSEA level. The inflammatory/chemokine signature we detect is more robust because it is driven by diverse, biologically coherent gene sets.

8.4 TIMP1-CD63 Not Replicated

The companion phylo analysis ([phylo_analysis/report_IVD_scrnaseq_analysis.md, Section 9](#)) identified TIMP1-CD63 loss as the dominant CCC change, alongside FN1-driven macrophage signaling gains. Our CCC analysis was dominated by HMGB1-CXCR4 axes. This discordance likely reflects differences in: (1) cell type resolution (12 vs. ~10 clusters), (2) CCC methodology (per-dataset vs. integrated), and (3) subsampling strategies. The TIMP1-CD63 finding remains biologically plausible (Vo et al., 2013) and warrants targeted investigation.

8.5 MDK (Midkine) as a Degeneration Marker

MDK was the most statistically significant DE gene in NP_mature_chondrocyte mild_vs_severe (log2FC=+2.72, padj=4.9x10⁻¹²). Midkine is a heparin-binding growth factor that promotes angiogenesis, inflammation, and cell survival. Its role in IVD degeneration has been minimally explored, though it has been implicated in osteoarthritis cartilage (Martel-Pelletier et al., 2020). MDK's extreme statistical significance (the lowest padj among all NP DE genes in this comparison) makes it a candidate biomarker and potential therapeutic target worthy of further investigation.

9. Limitations

Cross-study confounding: Condition and study are partially confounded, especially for herniated samples (only 2 studies). The NP_mature_chondrocyte healthy_vs_herniated comparison (4,316 DE genes) is explicitly flagged as study-confounded based on the presence of ribosomal protein batch artifacts among top genes. Within-study comparisons (mild_vs_severe) are prioritized throughout this manuscript.

Underpowered comparisons: 128 of 145 cell type x comparison combinations were skipped due to insufficient samples (< 3 per condition). The CEP compartment is entirely underpowered. Key genes like ADAMTS5, ACAN, and COL2A1 may fail to reach significance due to donor variability rather than absence of change.

Age-disease confound: In GSE230809 (the largest dataset, 24 samples), healthy donors are 21-27 years old and diseased are 37-73 years old. Age and degeneration effects cannot be fully separated.

Sex bias: GSE230809 is all-male. 30 of 78 samples have unknown sex. Sex-stratified analyses are not possible.

Culture-expanded cells: GSE242443 CEP cells are culture-expanded, which alters gene expression (particularly collagen ratios and surface markers).

No RNA velocity: Spliced/unspliced counts were not available from public deposits. RNA velocity would provide directional evidence for cell state transitions.

No SCENIC/GRN: Full SCENIC analysis (gene regulatory networks) was not performed due to computational requirements. TF activity was estimated from CollecTRI regulon overlap, which captures target enrichment but not regulatory network structure.

Composition analysis underpowered: No cell type proportion changes reached significance after FDR correction, though trends were biologically consistent (e.g., reduced NP notochordal cells, increased AF cells in degeneration).

Endothelial cluster contamination: The presence of ACAN and COL2A1 among endothelial DE genes suggests possible NP/AF cell misclassification into the endothelial cluster. Endothelial DE results should be interpreted cautiously.

CCC methodology: LIANA was run on per-dataset files (not integrated data), which avoids integration artifacts but fragments the analysis across datasets, potentially reducing power to detect consistent interaction changes.

10. Conclusion

This 12-dataset, 436,239-cell meta-analysis of human IVD degeneration reveals an inflammatory chemokine-dominated transcriptomic signature in severe NP degeneration, driven by coordinate upregulation of CXCL1/2/3 and TNF through NF- κ B signaling. This signature is accompanied by heat shock protein activation and mitochondrial dysfunction in AF cells, representing a tissue-wide stress response. Cell state trajectory analysis confirms that NP cells exist on a disease-associated continuum from notochordal to degenerative states, with pseudotime correlating with clinical disease severity.

The primary therapeutic opportunities emerging from this analysis are CXC chemokine blockade (via CXCR2 antagonism), TNF/NF- κ B inhibition, and HSP/proteostasis modulation — all supported by direct DE, pathway, and TF evidence from the data. Classical targets such as ADAMTS5 and TIMP1 remain valid based on extensive literature but were not independently confirmed in our powered comparisons.

Importantly, this analysis highlights the sensitivity of scRNA-seq meta-analysis results to methodological choices: integration method, comparison design, LFC shrinkage, and artifact awareness all substantially impact biological conclusions. Cross-study confounding and histone gene artifacts can produce apparently coherent but artifactual pathway signatures that propagate through enrichment analyses. Within-study comparisons, LFC shrinkage, and diverse gene set leading edges provide more robust biological insights.

11. References

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Analysis performed using a 10-module human-gated agentic pipeline. All code version-controlled. Random seed: 42. Package versions: Python 3.12, scanpy 1.11, scvi-tools 1.4.2, pyDESeq2, gseapy 1.1, decoupler 2.1, liana 1.7.

This is a computational analysis draft. All findings require experimental validation before clinical application.